ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 7 OF 14

Environment and lighting

Builder, environment artist, lighter

`scenario-roblox-environment`

Builds and lights Roblox maps from code: graybox, terrain, materials, foliage and moods, each checked by probes, voxel reads, image metrics and captures from fixed cameras.



Live capture, Studio 0.743: the village from the spawn camera in four looks, all under 5% crushed pixels.



Window-lit corridor with a PointLight [234, 255, 254], Range 50, as fake bounce. BONES, Advanced Lighting, 07:14.

USE IT WHEN

- Blocking out a level, landmarks and doorways, or turning parts into terrain
- A map looks flat, foggy, too dark or plastic
- Setting a day, sunset or night mood, or lighting an interior

WHAT THE EXPERTS INSIST ON

- Compose the map from where players spawn, and time the walk between landmarks. Gino Roy (Sawhorse Interactive)
- Cut axis-aligned holes with parts, not unions; each distinct union is its own mesh and collision asset. Kaiwala (Traitor Town map builder)
- Aim the sun first so light lands where it matters; move ColorCorrection in 0.05 steps. DatBoiEle (stylized builder, educator)
- Light window-lit rooms with Brightness 44, black ambients, and a PointLight near the window as fake bounce, Range 50. BONES (lead environment artist)
- Terrain uses a MaterialVariant only through an override, one per material: Grass and LeafyGrass separately. SmartyRBX (Roblox educator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Readiness check.** **Gate:** lighting style and file-only settings read as intended
2. **Graybox from parts.** **Gate:** clearance finds nothing; the spawn capture shows the first landmark
3. **Terrain from code.** **Gate:** spawn ground at design height; no grass through floors
4. **Openings and collision.** **Gate:** a character-sized probe and a ray pass every doorway
5. **Materials and scatter.** **Gate:** 0 broken material links; each terrain override reads back
6. **Light the moods.** **Gate:** metric bands met; contact sheet judged against the mood sentence
7. **Playtest and audit.** **Gate:** the character walks through a door; EnvAudit finds no error or warning

KEY NUMBERS

Doorways, halls and wall height	**10 studs at least**
Terrain region calls	**resolution 4, at most 4,194,304 voxels**
Daytime Atmosphere	**Density 0.3 to 0.4, Offset 0**
Dark corridor Atmosphere	**Density 0.495, Offset 0.25, Haze 5, Glare 0**
Light Range written by script	**clamped to 120 (live)**
Per-view budget on 2 GB phones	**500 draw calls, 500k triangles**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- Day looks: under 5% crushed pixels in every view, wider spread than neutral
- Outdoor night: crushed under 0.5 in every view, with a lit focal point near spawn
- Graybox clearance, material links and EnvAudit: zero findings at error or warn
- Contact sheet read by eye: mood matches, landmark contrasts, no near-black or plastic materials

HANDOFFS

Receives Meshes from scenario-roblox-assets, tileable PBR sets from scenario-textures, skyboxes from scenario-skyboxes.

Delivers A lit, audited map; playtest stats and EnvAudit counts to performance; mood and light positions to vfx-audio.

TOOLKIT `RBX_ENVIRONMENT.PY`

- `run()`: run code in Studio with the Graybox, Terrain, Looks kits
- `look_metrics()`: crushed, clipped, spread and warmth of a capture
- `lune_audit()`: audit the saved place file offline with Lune

MAIN EXPERT SOURCES

BONES (Urellity)
Lead environment artist, Advanced Lighting (2024)

DatBoiEle
Stylized builder and educator, The Ultimate Lighting Tutorial (2024)

Kaiwala
Builder of a Traitor Town map, Unions Are Bad For Building (2025)

Gino Roy (Heart_Hertz)
Art director, Sawhorse Interactive, Finding Your Art Direction, RDC 2024

STATUS

Tests 9 offline checks pass, 26 Lune tests

In app 17 live tests pass in Roblox Studio 0.743

G4 building and environment

without skills	8/15
with skills	15/15